Angio OMEX Experiment Report — angio_lchem_sweep_20261021_114558

3 combinations | 10, 10, 10 replicate(s) each | sim_time 100.0 MCS

Experiment Archive

Report OF the experiment archive **angio_lchem_sweep_scan.omex** (8,025 bytes) — the SED-ML/COMBINE description of the whole scan.

Property	Value
Parameter scan	lchem ∈ [100, 500, 1000]
Scan points	3
Replicates / point	10
Metrics compared	num_clusters, cell_density, largest_cluster_size, vegf_mean

Experiment OMEX Contents

File	Size (B)	Role
combo_data/lchem_100.csv	509	Per-combination replicate-mean metrics (CSV)
combo_data/lchem_1000.csv	492	Per-combination replicate-mean metrics (CSV)
combo_data/lchem_500.csv	432	Per-combination replicate-mean metrics (CSV)
manifest.xml	942	OMEX manifest (COMBINE archive index)
metadata.rdf	1,661	OMEX Metadata (provenance: creator, license, citation)
model/model_summary.json	2,831	Bundled model artifact (model/@source resolves here)
scan.sedml	13,550	SED-ML comparison (scan repeatedTask + plot2D per metric)
scan_metadata.json	9,789	Scan catalogue (points, combinations, metrics)

Reading the SED-ML (comparison)

plot2D (signal across combinations)	curves (= combinations)
num_clusters across combinations	3
cell_density across combinations	3
largest_cluster_size across combinations	3
vegf_mean across combinations	3

Validation of the Experiment Archive

Layer	Result
Built-in SED-ML	PASS
Built-in manifest	PASS
biosimulators-utils	PASS (0 errors, 1 warning group)

Provenance (OMEX Metadata)

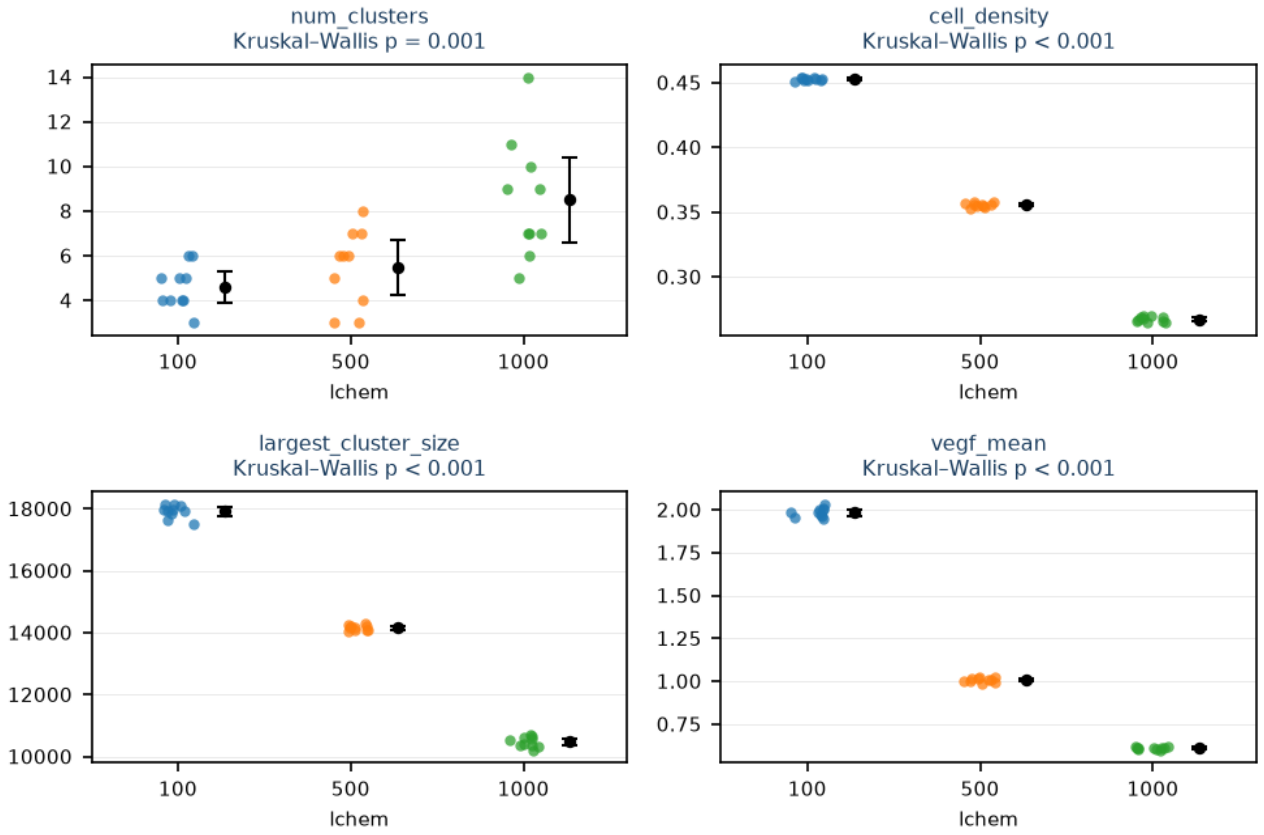
Field	Value
Creator (experimenter)	Joel Vanin
Created	2026-10-21T19:04:52.879144
Modified	2026-10-21T19:04:52.879144
License	https://spdx.org/licenses/CC-BY-4.0.html
Citation	Merks, R.M.H., Brodsky, S.V., Goligorksy, M.S., Newman, S.A., Glazier, J.A. (2006). Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling. Developmental Biology 289(1), 44-54. https://doi.org/10.1016/j.ydbio.2005.10.003
Keywords	angiogenesis, parameter scan, cellular Potts model, VEGF

Summary Statistics (final step, per replicate)

Each metric is read at the final step of every replicate, then summarised across the replicates of a combination as mean ± SD with the 95% confidence interval of the mean (t distribution) below it. The last column is a Kruskal–Wallis test of whether the metric differs across the combinations.

Metric / lchem	100 (n=10)	500 (n=10)	1000 (n=10)	Kruskal–Wallis p
num_clusters	4.60 ± 0.97 [3.91, 5.29]	5.50 ± 1.72 [4.27, 6.73]	8.50 ± 2.68 [6.58, 10.42]	0.001
cell_density	0.453 ± 0.001 [0.452, 0.454]	0.356 ± 0.002 [0.354, 0.357]	0.267 ± 0.002 [0.266, 0.269]	<0.001
largest_cluster_size	17,914 ± 211 [17,764, 18,065]	14,166 ± 84 [14,106, 14,226]	10,477 ± 161 [10,362, 10,593]	<0.001
vegf_mean	1.98 ± 0.03 [1.97, 2.00]	1.01 ± 0.01 [1.00, 1.02]	0.612 ± 0.007 [0.607, 0.617]	<0.001

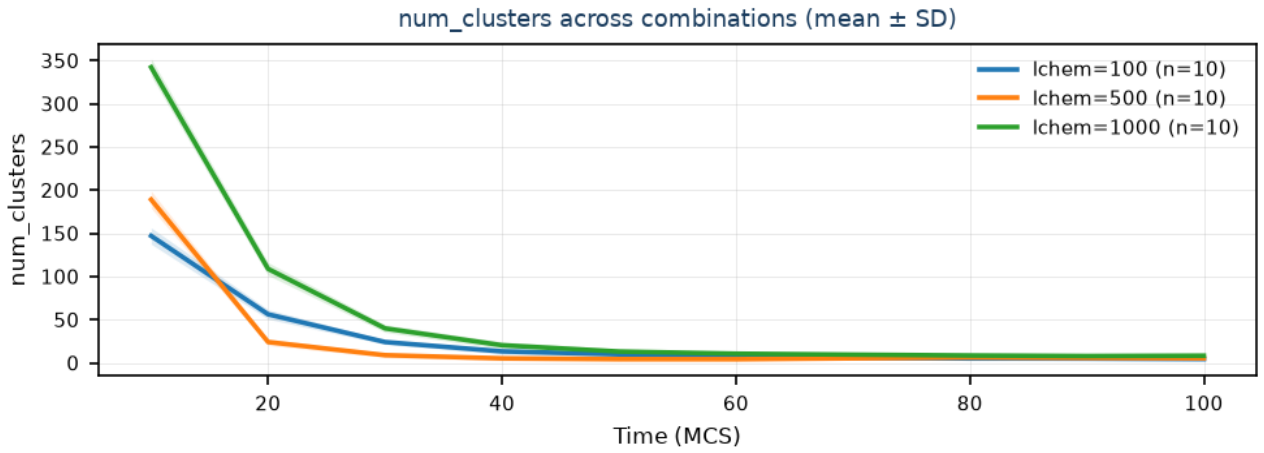
With few replicates per combination the tests have little power and are not corrected for the number of metrics; read them as descriptive.

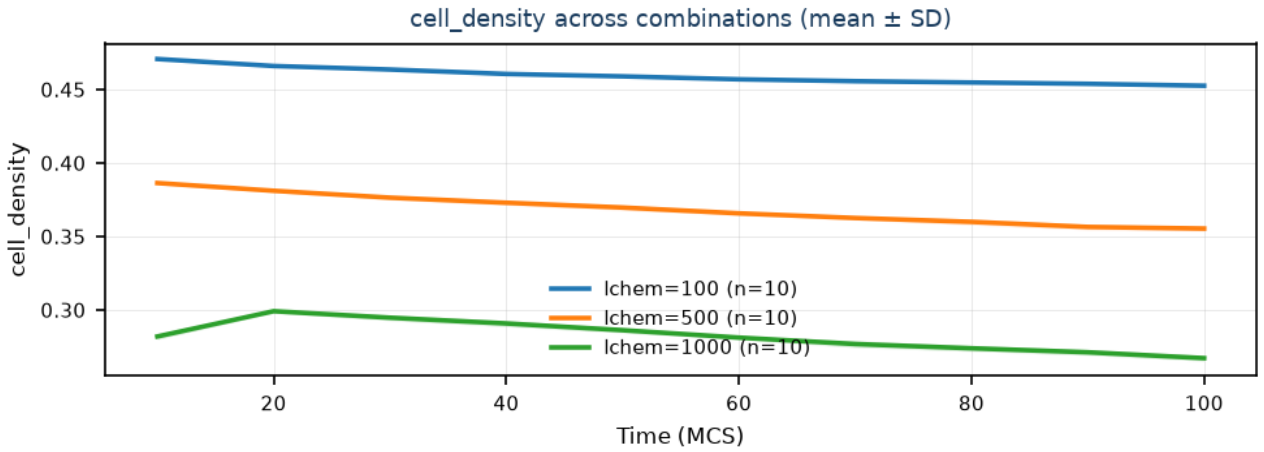


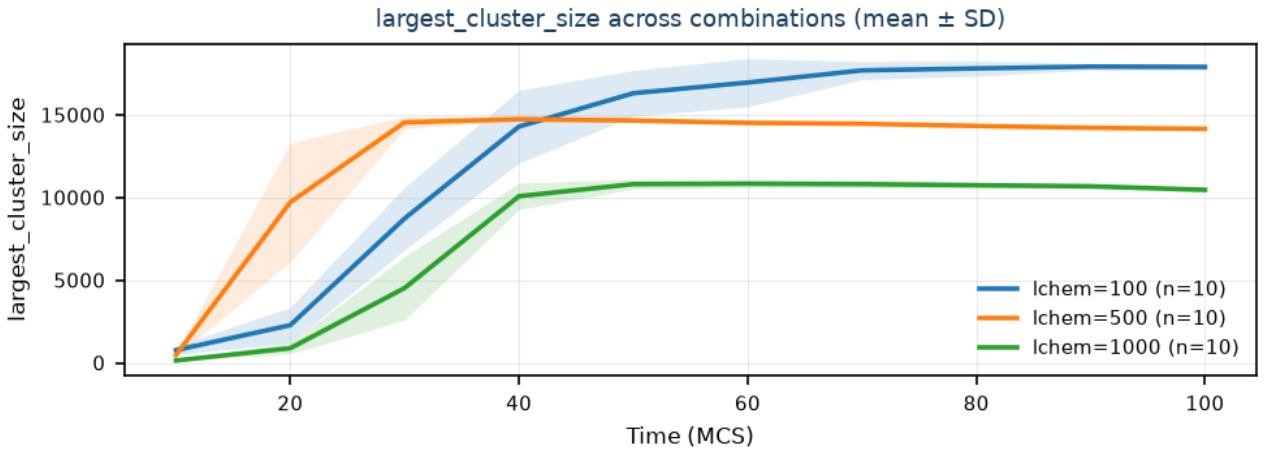
Final-step value of each metric by combination: each dot is one replicate; the black point and bar are the mean and its 95% confidence interval.

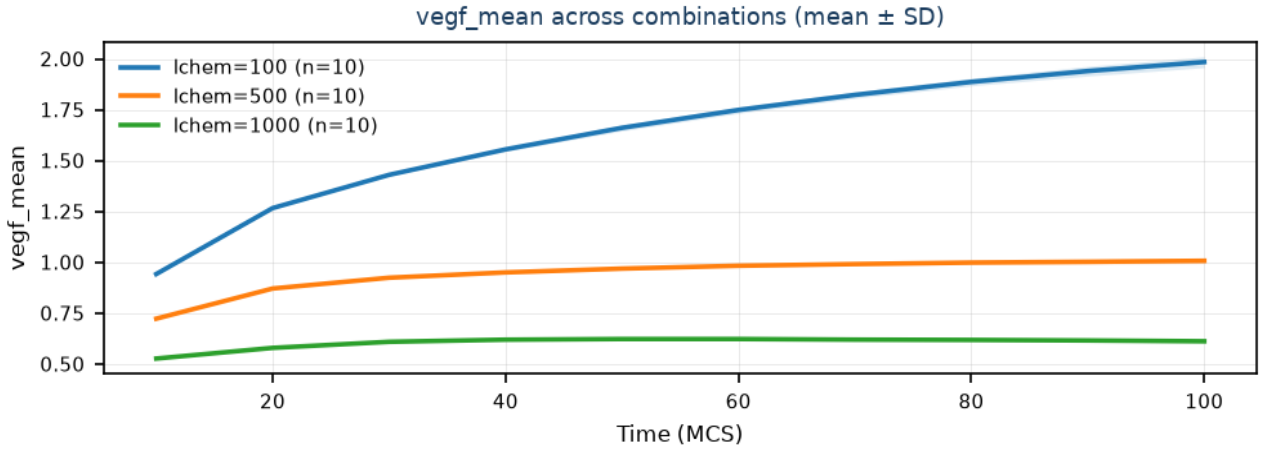
Metric Comparisons (one line per combination)

Each line is a combination's across-replicate mean and the shaded band ±1 SD across its replicates; the legend gives the number of replicates n. Time in MCS.

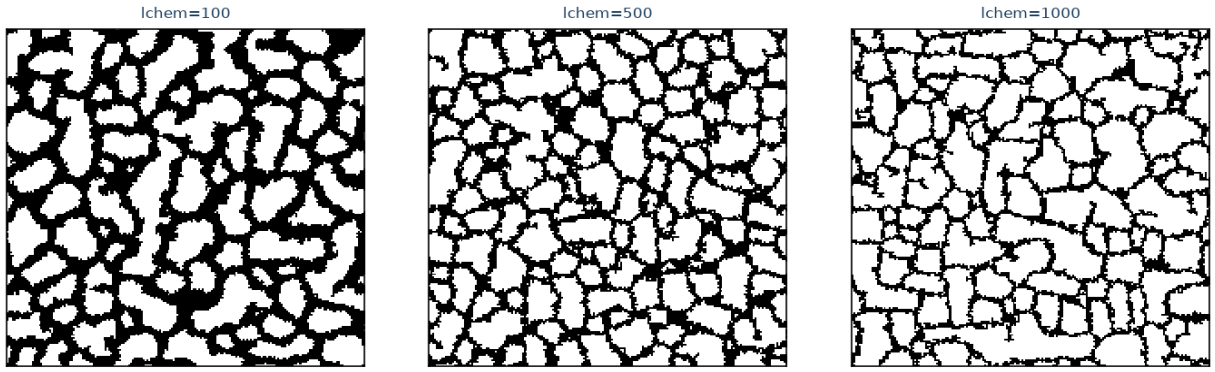








Final-step EC networks (one per combination)



Final-step VEGF fields (one per combination)

